



Report 5

Enterprise Network Design Using Core, Distribution, and Access Layers
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July 23, 2026

**NETWORK SECURITY –
BATCH 2**



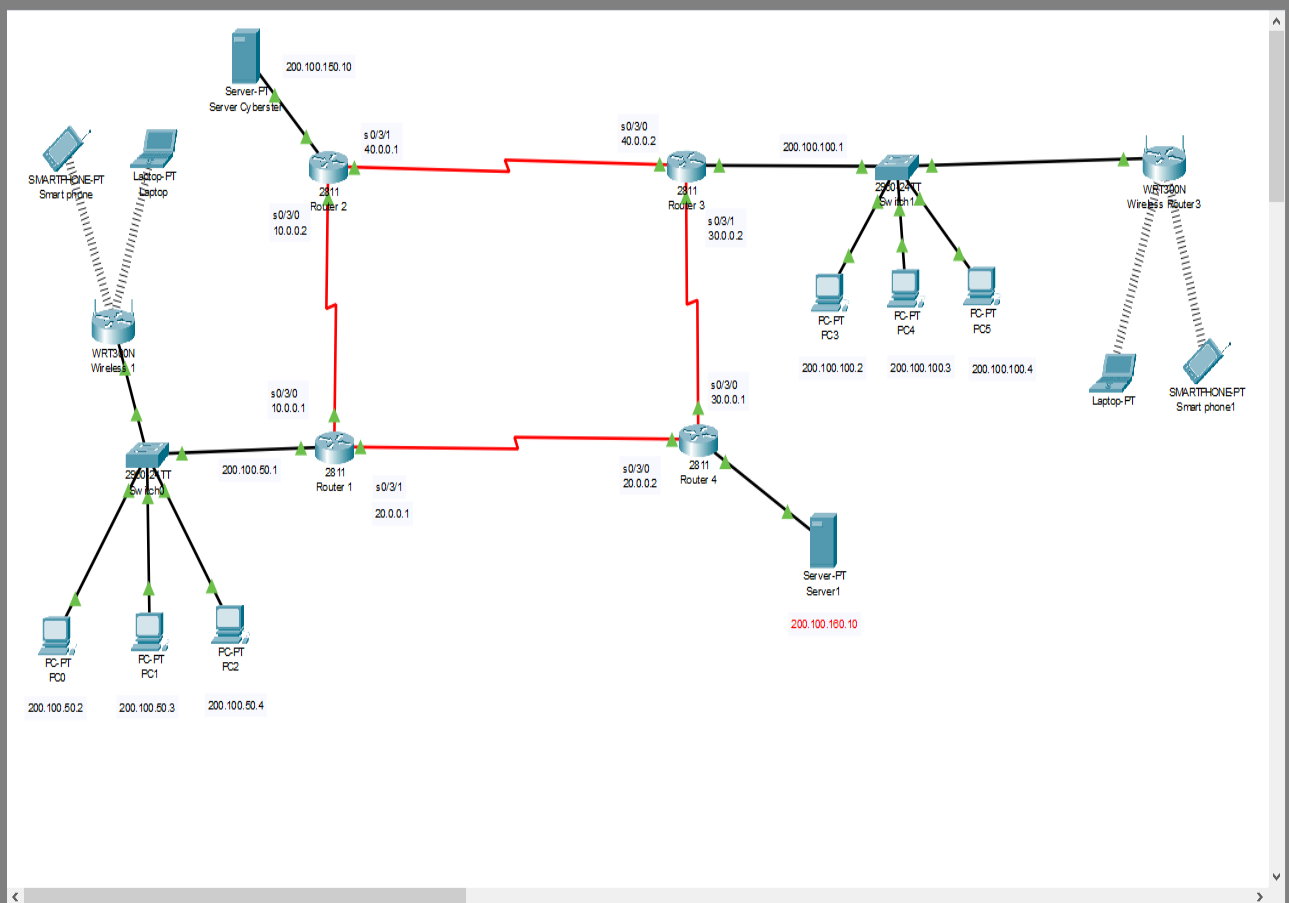
Objective:

The objective of this task was to design and implement an enterprise network in Cisco Packet Tracer using static routing. The network follows the three-tier hierarchical model consisting of the Core, Distribution, and Access layers. The topology includes routers, switches, servers, wired hosts, and wireless devices. End-to-end connectivity between all networks was verified.

Task Assigned:

- Design an enterprise network using routers and switches.
- Implement the Core, Distribution, and Access layers.
- Configure static routing between all routers.
- Connect wired and wireless end devices.
- Configure two servers on different networks.
- Verify complete connectivity between all devices.

Network Topology:



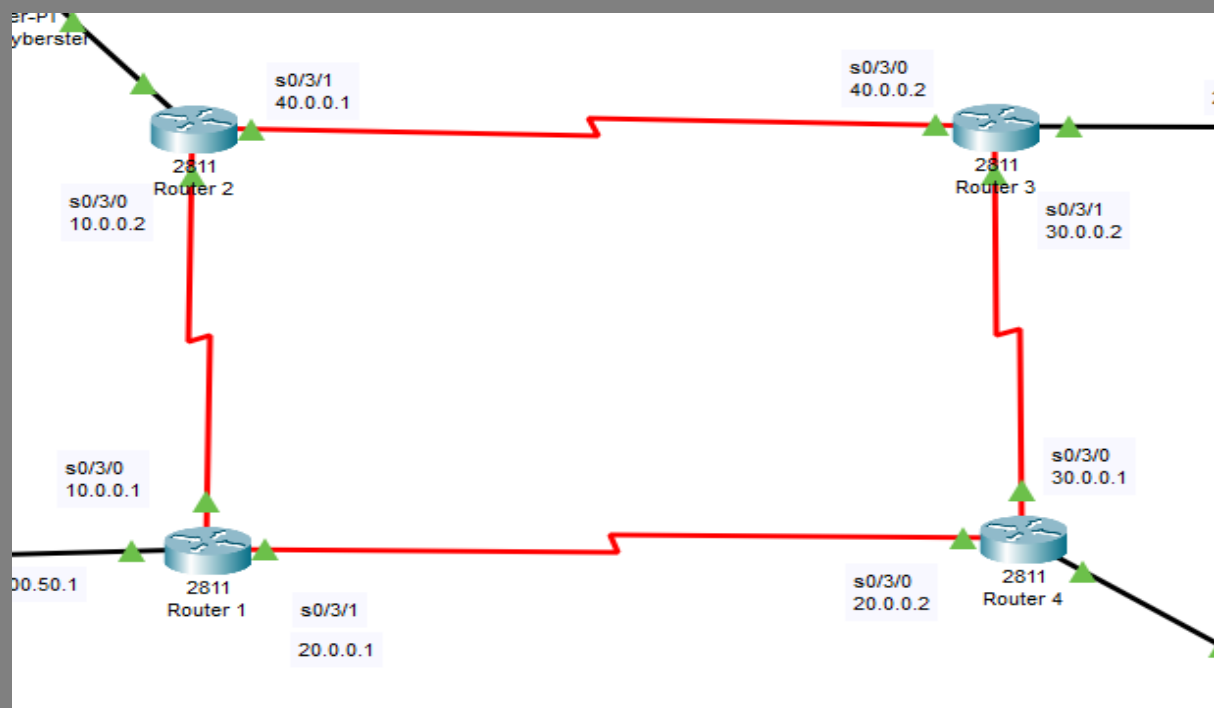
Network Devices:

Device	Quantity
Cisco 2811 Router	4
Cisco 2960 Switch	2
PCs	5
Wireless Router / Access Point	2
Laptop	2
Smartphone	2
Servers	2

Network Design:

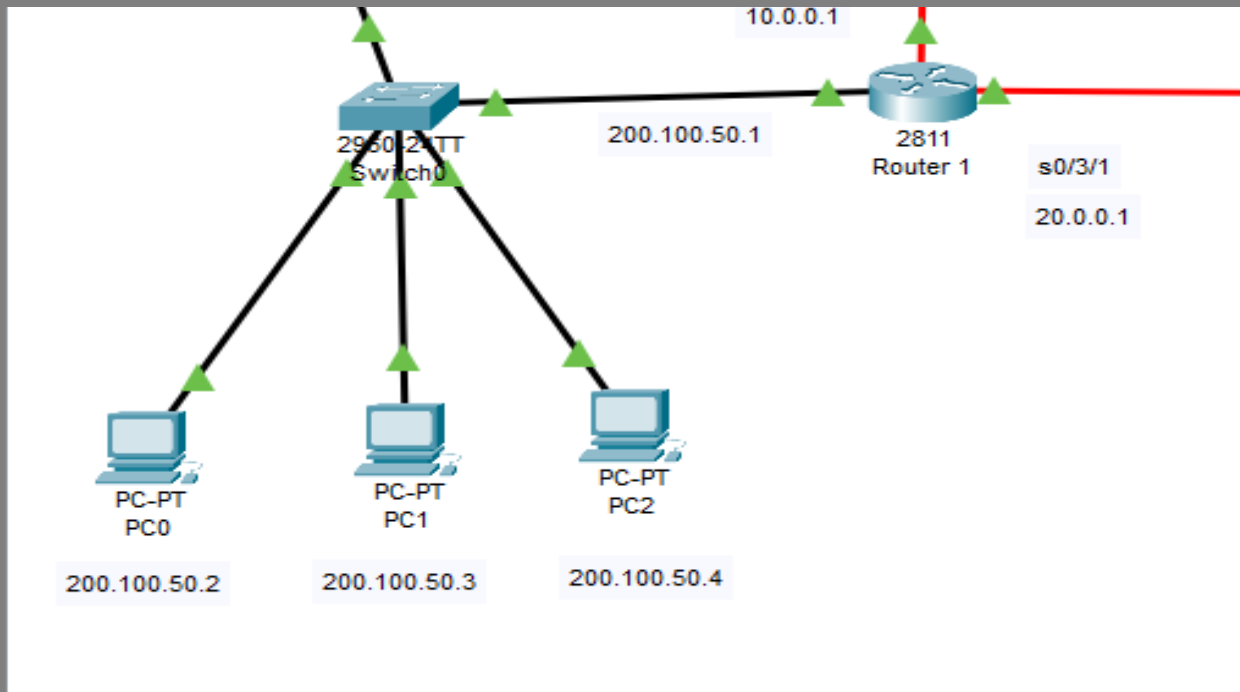
Core Layer

The Core Layer consists of four Cisco 2811 routers connected using serial links. These routers are responsible for forwarding traffic between different networks using static routing.

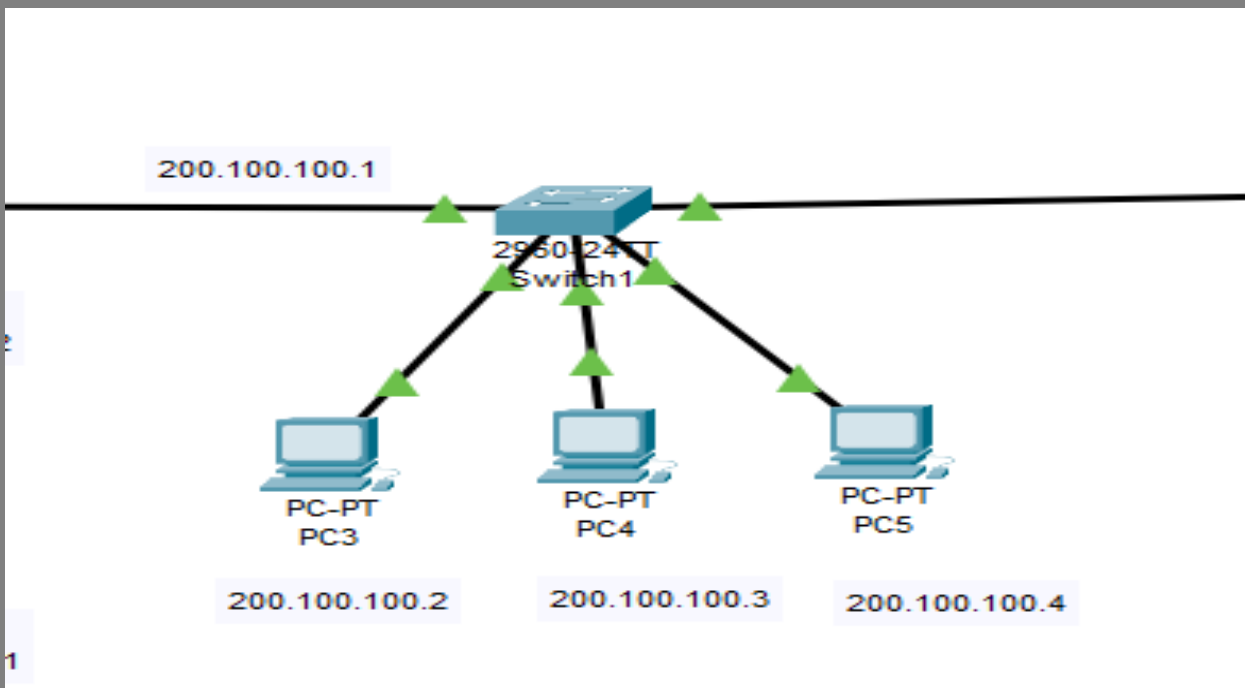


Distribution Layer:

Cisco 2960 switches were connected to the routers. The switches aggregate traffic from access devices and forward it to the core routers.

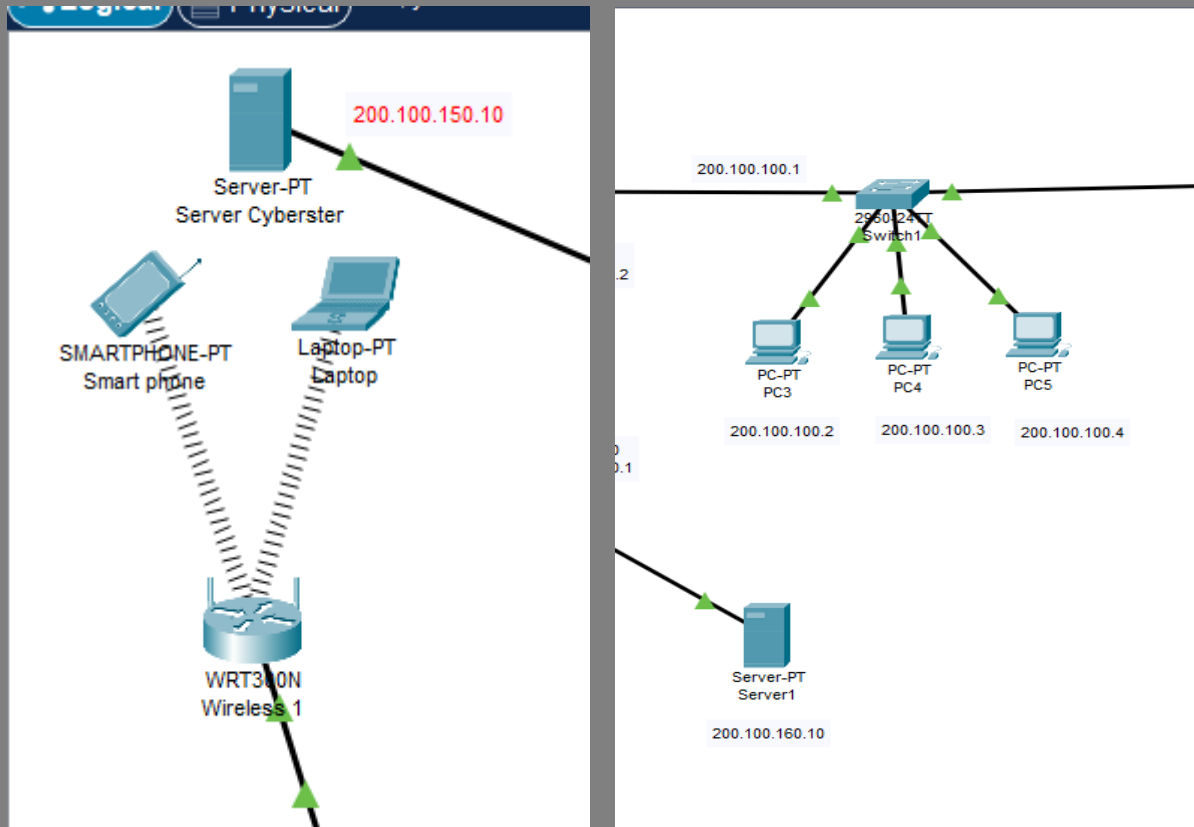


Switch 2:



Access Layer:

The Access Layer contains wired PCs, wireless devices, and servers connected to the distribution switches and routers.



IP Addressing:

Network	Address
LAN 1	200.100.50.0/24
LAN 2	200.100.100.0/24
Server Network 1	200.100.150.0/24
Server Network 2	200.100.160.0/24
Router Links	10.0.0.0, 20.0.0.0, 30.0.0.0, 40.0.0.0

Verification Commands:

R1: show ip int brief

```
R1#
R1#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.50.1    YES manual up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial10/3/0             10.0.0.1        YES manual up          up
Serial10/3/1             20.0.0.1        YES manual up          up
Vlan1                    unassigned      YES unset   administratively down down
R1#
```

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R1: Show ip route

```
Vlan1                    unassigned      YES unset   administratively down down
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C       10.0.0.0/30 is directly connected, Serial10/3/0
   L       10.0.0.1/32 is directly connected, Serial10/3/0
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C       20.0.0.0/30 is directly connected, Serial10/3/1
   L       20.0.0.1/32 is directly connected, Serial10/3/1
  30.0.0.0/30 is subnetted, 1 subnets
   S       30.0.0.0/30 [1/0] via 20.0.0.2
  40.0.0.0/30 is subnetted, 1 subnets
   S       40.0.0.0/30 [1/0] via 10.0.0.2
 200.100.50.0/24 is variably subnetted, 2 subnets, 2 masks
   C       200.100.50.0/24 is directly connected, FastEthernet0/0
   L       200.100.50.1/32 is directly connected, FastEthernet0/0
   S       200.100.100.0/24 [1/0] via 10.0.0.2
   S       200.100.150.0/24 [1/0] via 10.0.0.2
   S       200.100.160.0/24 [1/0] via 20.0.0.2
R1#
```

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R1: show running-config:

```
!
!
!
interface FastEthernet0/0
 ip address 200.100.50.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 10.0.0.1 255.255.255.252
!
interface Serial0/3/1
 ip address 20.0.0.1 255.255.255.252
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 200.100.100.0 255.255.255.0 10.0.0.2
ip route 30.0.0.0 255.255.255.252 20.0.0.2
ip route 40.0.0.0 255.255.255.252 10.0.0.2
ip route 200.100.150.0 255.255.255.0 10.0.0.2
ip route 200.100.160.0 255.255.255.0 20.0.0.2
!
ip flow-export version 9
!
!
!
```

R2: show ipint brief

```
R2#
R2#
R2#
R2#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.150.1   YES manual up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/3/0               10.0.0.2        YES manual up          up
Serial0/3/1               40.0.0.1        YES manual up          up
Vlan1                    unassigned      YES unset   administratively down down
R2#
```

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R2: Show IP route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
C       40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/30 is directly connected, Serial0/3/1
L       40.0.0.1/32 is directly connected, Serial0/3/1
S       200.100.50.0/24 [1/0] via 10.0.0.1
S       200.100.100.0/24 [1/0] via 40.0.0.2
C       200.100.150.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.100.150.0/24 is directly connected, FastEthernet0/0
L       200.100.150.1/32 is directly connected, FastEthernet0/0
S       200.100.160.0/24 [1/0] via 40.0.0.2
```

R2:Show running-config

```
!
interface FastEthernet0/0
 ip address 200.100.150.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 10.0.0.2 255.255.255.252
 clock rate 64000
!
interface Serial0/3/1
 ip address 40.0.0.1 255.255.255.252
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 200.100.50.0 255.255.255.0 10.0.0.1
ip route 200.100.100.0 255.255.255.0 40.0.0.2
ip route 200.100.150.0 255.255.255.0 10.0.0.1
ip route 200.100.160.0 255.255.255.0 40.0.0.2
!
ip flow-export version 9
!
!
```

R3: Show ip int brief

```
R3>en
R3#
R3#show ip int brief
Interface                IP-Address      OK? Method Status              Protocol
FastEthernet0/0          200.100.100.1   YES manual up                    up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/3/0               40.0.0.2        YES manual up                    up
Serial0/3/1               30.0.0.2        YES manual up                    up
Vlan1                     unassigned      YES unset  administratively down down
R3#
```

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R3: Show ip Route

```
Gateway of last resort is not set

 20.0.0.0/30 is subnetted, 1 subnets
S    20.0.0.0/30 [1/0] via 30.0.0.1
 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/30 is directly connected, Serial0/3/1
L    30.0.0.2/32 is directly connected, Serial0/3/1
 40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/30 is directly connected, Serial0/3/0
L    40.0.0.2/32 is directly connected, Serial0/3/0
S    200.100.50.0/24 [1/0] via 40.0.0.1
 200.100.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.100.100.0/24 is directly connected, FastEthernet0/0
L    200.100.100.1/32 is directly connected, FastEthernet0/0
S    200.100.150.0/24 [1/0] via 40.0.0.1
S    200.100.160.0/24 [1/0] via 30.0.0.1

R3#
R3#
```


R3: Show Running-Config

```
!
interface FastEthernet0/0
ip address 200.100.100.1 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/3/0
ip address 40.0.0.2 255.255.255.252
clock rate 2000000
!
interface Serial0/3/1
ip address 30.0.0.2 255.255.255.252
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 200.100.50.0 255.255.255.0 40.0.0.1
ip route 20.0.0.0 255.255.255.252 30.0.0.1
ip route 200.100.150.0 255.255.255.0 40.0.0.1
ip route 200.100.100.0 255.255.255.0 30.0.0.1
ip route 200.100.160.0 255.255.255.0 30.0.0.1
!
ip flow-export version 9
!
```

R4: show ip int brief

```
R4#
R4#
R4#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.160.1   YES manual up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/3/0               30.0.0.1        YES manual up          up
Serial0/3/1               20.0.0.2        YES manual up          up
Vlan1                     unassigned      YES unset   administratively down down
```

R4: show ip route

```
1 - periodic downloaded route table

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/30 is directly connected, Serial0/3/1
L    20.0.0.2/32 is directly connected, Serial0/3/1
  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/30 is directly connected, Serial0/3/0
L    30.0.0.1/32 is directly connected, Serial0/3/0
S    200.100.50.0/24 [1/0] via 20.0.0.1
S    200.100.100.0/24 [1/0] via 30.0.0.2
S    200.100.150.0/24 [1/0] via 30.0.0.2
  200.100.160.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.100.160.0/24 is directly connected, FastEthernet0/0
L    200.100.160.1/32 is directly connected, FastEthernet0/0
```

R4: show running-config

```
!
interface FastEthernet0/0
 ip address 200.100.160.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 30.0.0.1 255.255.255.252
 clock rate 64000
!
interface Serial0/3/1
 ip address 20.0.0.2 255.255.255.252
 clock rate 64000
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 200.100.160.0 255.255.255.0 30.0.0.2
ip route 200.100.50.0 255.255.255.0 20.0.0.1
ip route 200.100.100.0 255.255.255.0 30.0.0.2
ip route 200.100.150.0 255.255.255.0 30.0.0.2
!
ip flow-export version 9
!
!
```

Testing Results:

PC 0 To PC 3 Ping

```
C:\>ping 200.100.100.2

Pinging 200.100.100.2 with 32 bytes of data:

Reply from 200.100.100.2: bytes=32 time=37ms TTL=125
Reply from 200.100.100.2: bytes=32 time=32ms TTL=125
Reply from 200.100.100.2: bytes=32 time=25ms TTL=125
Reply from 200.100.100.2: bytes=32 time=31ms TTL=125

Ping statistics for 200.100.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 37ms, Average = 31ms

C:\>|
```

PC 3 To PC 0 ping

```
C:\>ping 200.100.50.2

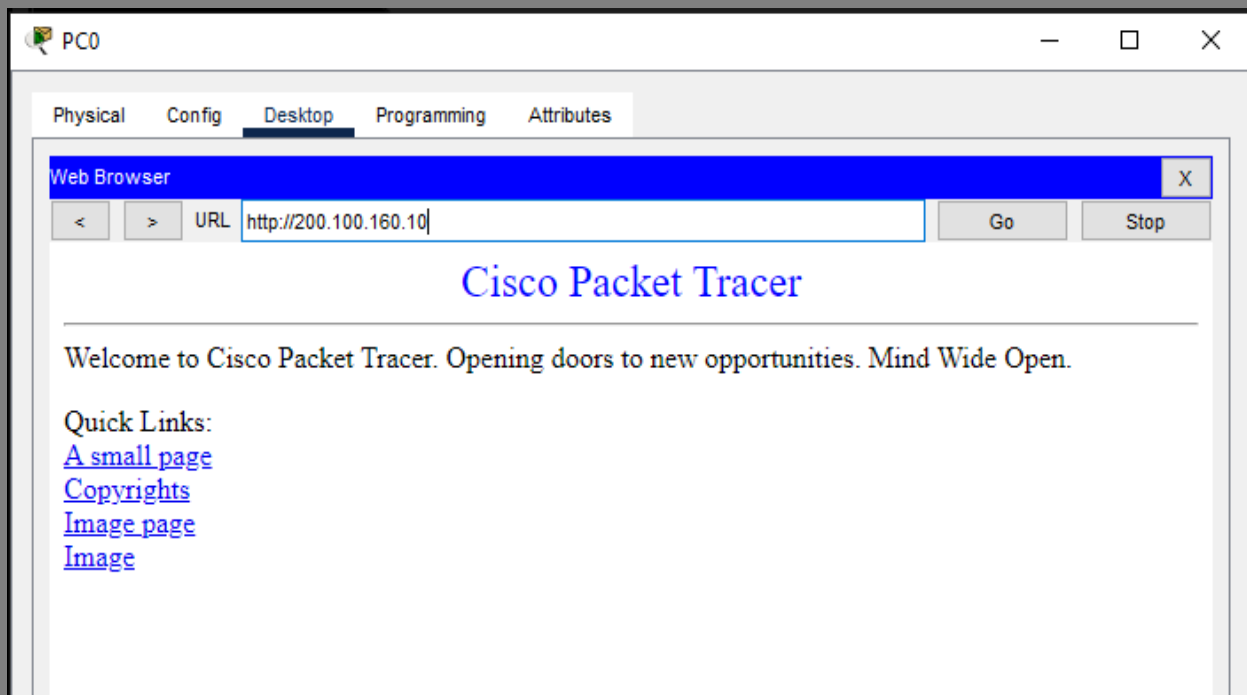
Pinging 200.100.50.2 with 32 bytes of data:

Reply from 200.100.50.2: bytes=32 time=40ms TTL=125
Reply from 200.100.50.2: bytes=32 time=31ms TTL=125
Reply from 200.100.50.2: bytes=32 time=22ms TTL=125
Reply from 200.100.50.2: bytes=32 time=22ms TTL=125

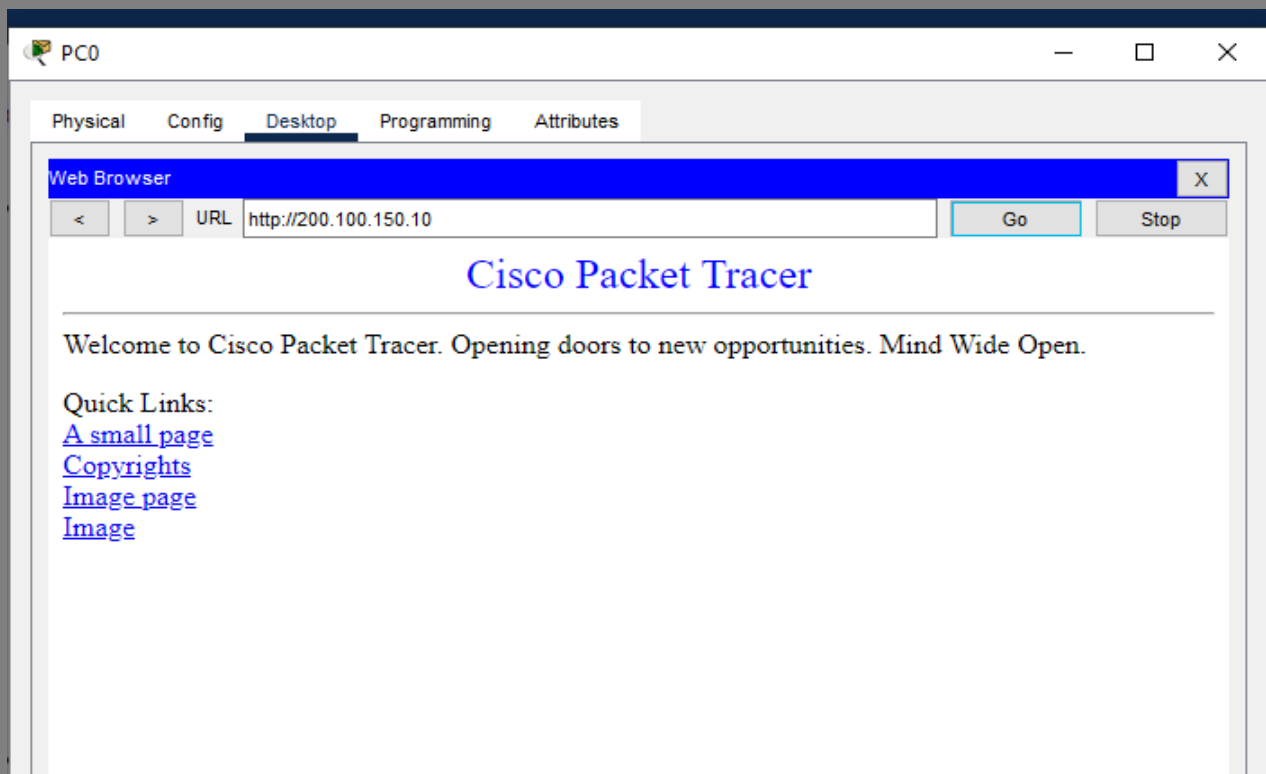
Ping statistics for 200.100.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 40ms, Average = 28ms

C:\>|
```

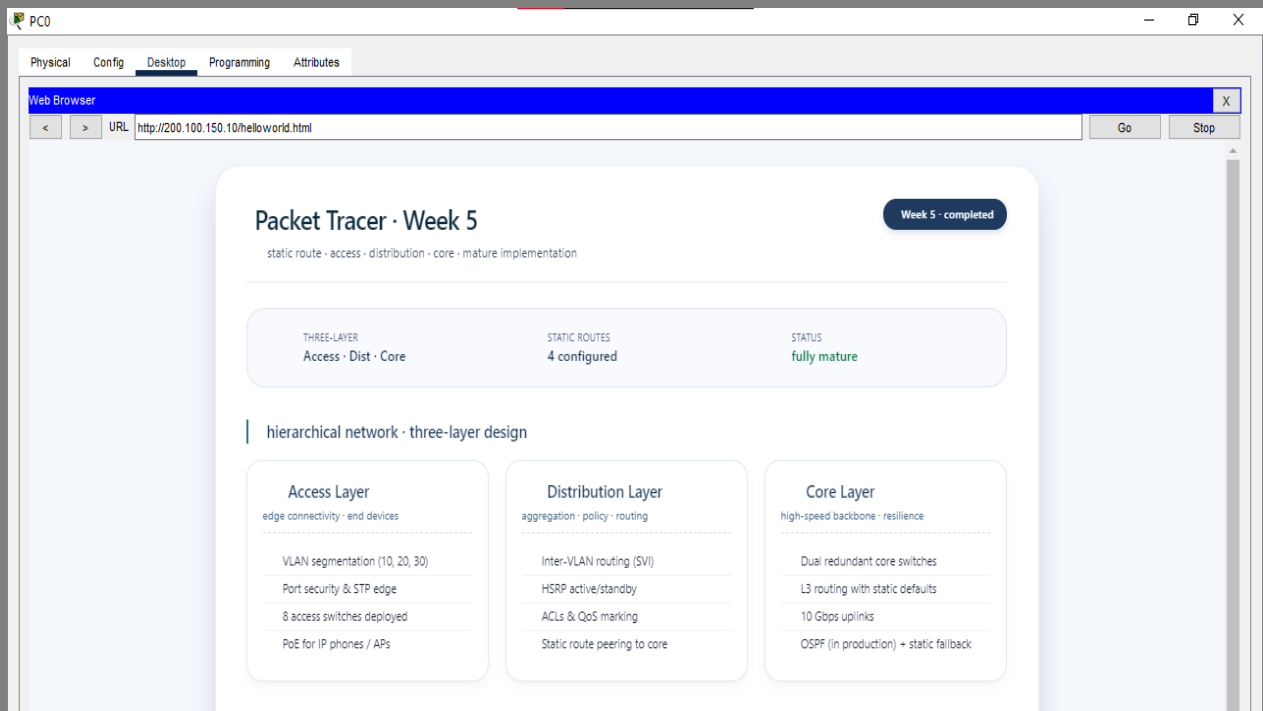
PC 0 To Server 2 Ping:



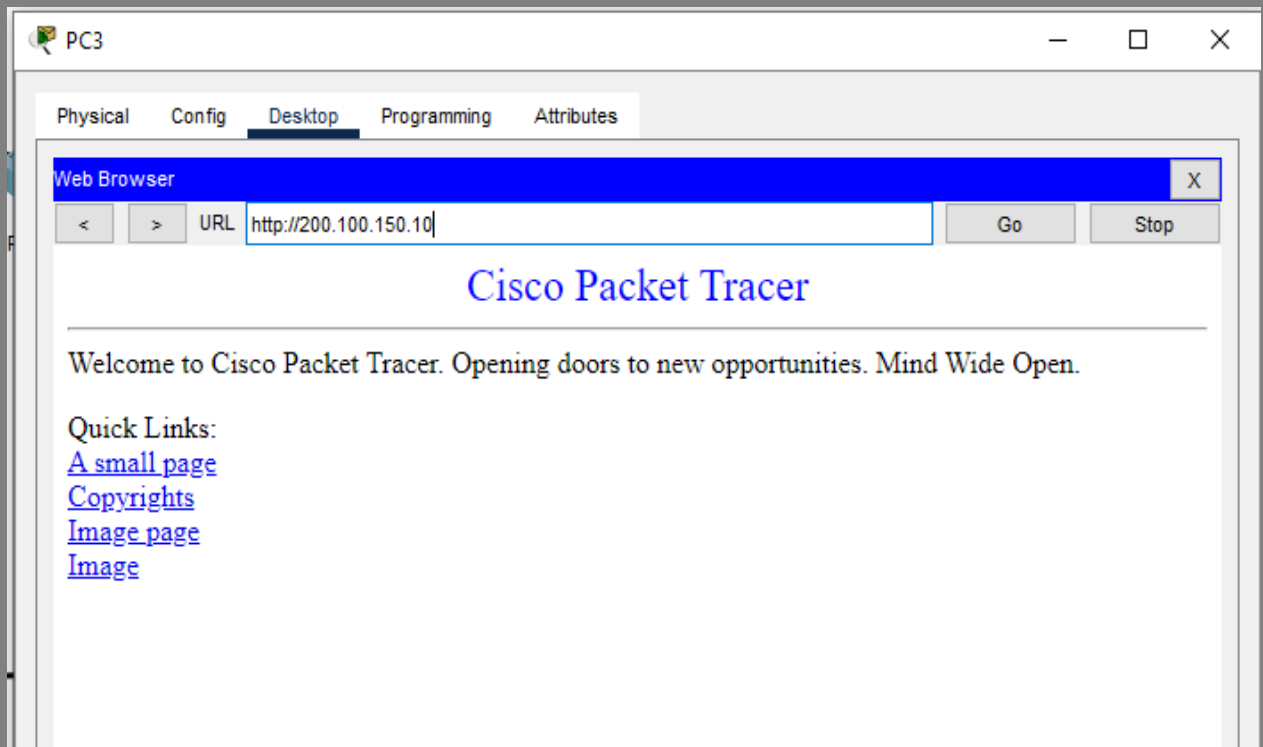
PC 0 To Server 2 Ping:



PC 0 To Server 1:



PC 3 To Server 1 Ping:



Laptop To Wireless Router Ping:

```
C:\>ping 200.100.50.10

Pinging 200.100.50.10 with 32 bytes of data:

Reply from 200.100.50.10: bytes=32 time=76ms TTL=255
Reply from 200.100.50.10: bytes=32 time=40ms TTL=255
Reply from 200.100.50.10: bytes=32 time=25ms TTL=255
Reply from 200.100.50.10: bytes=32 time=49ms TTL=255

Ping statistics for 200.100.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 76ms, Average = 47ms

C:\>
```

Learning Experience:

This task improved my understanding of enterprise network design using the three-tier architecture. I learned how to configure static routing, assign IP addressing, connect servers and wireless devices, and troubleshoot routing issues. It also strengthened my practical skills in Cisco Packet Tracer and network verification

Challenges Faced:

- Configuring static routes correctly on all routers.
- Troubleshooting missing routes causing unreachable networks.
- Configuring wireless connectivity.
- Verifying communication between servers and remote LANs.

Conclusion:

The enterprise network was successfully designed and implemented using Cisco Packet Tracer. The topology followed the Core, Distribution, and Access layer architecture. Static routing enabled communication between all networks, including wired devices, wireless clients, and servers. All connectivity tests were successful, demonstrating a functional and scalable enterprise network.